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Motion Planning for the Heat Equation

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Introduction

We are concerned with the null and approximate controllability of the heat equation in one dimension when a boundary control acts on the system. We provide an explicit and very regular control by using the so-called flatness approach, which consists in parametrizing the solution and the control by the derivatives of a "flat output". Choosing a suitable trajectory for this flat output yields an explicit series for the control that achieves controllability results.

The Flatness Method

- Introduced in the nineties by M. Fliess, J. Lévine, P. Martin and P. Rouchon[1].
- Consider a smooth control system

(1) $$z'(t) = f(z(t), u(t)), \quad z(t) \in \mathbb{R}^n, \quad u(t) \in \mathbb{R}^m$$

with an output $y(t) \in \mathbb{R}^m$ such that $y = h\left(z, u, u', \dots, u^{(r)}\right)$ for some smooth function h.

- We say the system (**??**) is flat if

$$z = g\left(y, y', \cdots, y^{(p)}\right), \quad \text{and} \quad u = \varphi\left(y, y', \cdots, y^{(q)}\right),$$

where g and φ are some smooth functions.

An Illustrative Examples

- Consider the linearized inverted pendulum

(2) $$\begin{cases} z_1'(t) = z_2(t), \\ z_2'(t) = u(t), \end{cases}$$

where $z = (z_1, z_2) \in \mathbb{R}^2$ is the state and $u(t) \in \mathbb{R}$ is the control.

- Choose an output $y(t) = z_1(t), \quad t \geq 0$.
- $z_1 = y$, $z_2 = y'$ and $u = y'' \implies$ The output y is flat.
- Choose the initial and final state $z_0 = (0, 0)$ and $z_T = (1, 0)$. We have to find $y \in C^\infty([0, T], \mathbb{R})$ such that

$$y(0) = 0, \quad y'(0) = 0, \quad y(T) = 1, \quad y'(T) = 0.$$

- One suitable choice is

$$y(t) = \frac{t^2(2T - t)^2}{T^4}, \quad t \in [0, T].$$

The Heat Equation in 1d

The heat equation in the interval $(0, 1)$ is

(3) $$\begin{cases} \partial_t \theta(t, x) = \partial_{xx}\theta(t, x), & x \in (0, 1), \\ \partial_x \theta(t, 0) = 0, \quad \theta_x(t, 1) = u(t), & t \in (0, T), \\ \theta(0, x) = \theta_0(x), & x \in (0, 1). \end{cases}$$

- θ is the temperature along the rod, θ_0 is the initial temperature and
- u is the control input.

Goal: Study the null and approximate controllability of the heat equation (3).

Gevrey Class of Functions

- Introduced by Maurice Gevrey in 1918.
- Let $\Omega \subset \mathbb{R}$ be an interval and $\alpha > 0$. Then,

$$\mathcal{C}^\omega(\Omega) \subset \mathcal{G}^\alpha(\Omega) \subset \mathcal{C}^\infty(\Omega).$$

- Let $\alpha > 0$. A function $y \in \mathcal{C}^\infty[0, T]$ is said to be Gevrey of order α, denoted $y \in \mathcal{G}^\alpha[0, T]$, if

$$\exists M, R > 0, \text{ s.t. } \forall m \in \mathbb{N} \cup \{0\}, \ \sup_{t \in [0,T]} \left|y^{(m)}(t)\right| \leq M \frac{(m!)^\alpha}{R^m}.$$

- Gevrey functions of order 1 are analytic and entire if $0 < \alpha < 1$.
- A Gevrey function of order α is also of order β for any $\beta \geq \alpha$.
- Gevrey functions of order $\alpha > 1$ may be constant on an open set without being constant everywhere.
- Let $\gamma > 0$. The functions

$$\varphi_\gamma(t) := \begin{cases} \exp\left(\frac{-1}{(1-t)^\gamma}\right), & t \in (0, 1), \\ 0, & t \notin (0, 1), \end{cases} \quad \Phi_\gamma(t) := \begin{cases} \frac{\int_0^t \varphi_\gamma(\tau) d\tau}{\int_0^1 \varphi_\gamma(\tau) d\tau}, & t \in (0, 1), \\ 0, & t \leq 0, \\ 1, & t \geq 1, \end{cases}$$

are Gevrey of order $1 + \frac{1}{\gamma}$.

- Let $\alpha, \beta > 0$. A function $\theta \in \mathcal{C}^\infty([0, T] \times [0, 1])$ is said to be Gevrey of order α in t and β in x, denoted $\theta \in \mathcal{G}^{\alpha,\beta}([0, T] \times [0, 1])$, if $\exists M, R, S > 0$, such that $\forall m, n \in \mathbb{N} \cup \{0\}$

$$\sup_{(t,x) \in [0,T] \times [0,1]} \left|\frac{\partial^{m+n}\theta}{\partial t^m \partial x^n}(t, x)\right| \leq M \frac{(m!)^\alpha (n!)^\beta}{R^m S^n}.$$

The Heat Equation is "Flat"

- We seek $\theta(t, x) = \sum_{k \geq 0} a_k(t) \frac{x^k}{k!}$ and define an output $y(t) := \theta(t, 0)$.
- If θ solves the heat equation (3), we find

$$a_{2k}(t) = y^{(k)}(t), \quad a_{2k+1}(t) = 0, \quad \forall k \geq 0.$$

- The formal solution is

$$\theta(t, x) = \sum_{k \geq 0} y^{(k)}(t) \frac{x^{2k}}{(2k)!}.$$

- The formal control is

$$u(t) = \sum_{k \geq 0} \frac{y^{(k)}(t)}{(2k - 1)!}.$$

- If $y \in \mathcal{G}^\alpha[0, T]$ with $0 < \alpha < 2$, the formal solution $\theta \in \mathcal{G}^{\alpha,1}([0, T] \times [0, 1])$. In particular, the formal control $u \in \mathcal{G}^\alpha[0, T]$.

Controllability of Heat Equation

Theorem 1 (Null Controllability)[3].
Let $\theta_0 \in L^2(0, 1)$. Pick $\tau \in (0, T)$ and $s \in (1, 2)$. There exists $y \in \mathcal{G}^s[\tau, T]$ such that, setting

$$u(t) = \begin{cases} 0, & \text{if } 0 \leq t \leq \tau, \\ \sum_{k \geq 0} \frac{y^{(k)}(t)}{(2k-1)!}, & \text{if } \tau < t \leq T, \end{cases}$$

the solution θ of (3) satisfies $\theta(T, \cdot) = 0$ on $(0, 1)$.

Theorem 2 (Approximate Controllability)[2].
Let $\theta_0, \theta_T \in L^2(0, 1)$ and $s \in (1, 2)$. For any given $\epsilon > 0$, there exists $y_\epsilon \in \mathcal{G}^s[\tau, T]$ such that, setting

$$u_\epsilon(t) = \sum_{k \geq 0} \frac{y_\epsilon^{(k)}(t)}{(2k - 1)!}, \quad \text{if } 0 < t \leq T,$$

the solution θ_ϵ of (3) satisfies $\|\theta_\epsilon(T, \cdot) - \theta_T(\cdot)\| < \epsilon$.

References

1. Fliess, Michel; Lévine, Jean; Martin, Philippe; Rouchon, Pierre. *Flatness and defect of non-linear systems: introductory theory and examples*. Internat. J. Control 61 (1995).
2. Béatrice Laroche, Philippe Martin, and Pierre Rouchon. *Motion planning for the heat equation*. volume 10, pages 629-643, (2000).
3. Martin, Philippe; Rosier, Lionel; Rouchon, Pierre. *Null controllability of the heat equation using flatness*. Automatica J. IFAC 50 (2014).

Remarks. The flatness method can be useful in various ways. For instance, proving controllability of ODEs, Heat Equation, Korteweg-de Vries (KdV) equation, Schrödinger equation etc. The control is explicit (more regular) and the trajectories are easy to approximate. This method gives characterization of Reachable Sets for some PDEs (heat equation, KdV equation). This method is very helpful to prove the existence of the kernels that occur in the Backstepping or in the Transmutation Method.

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